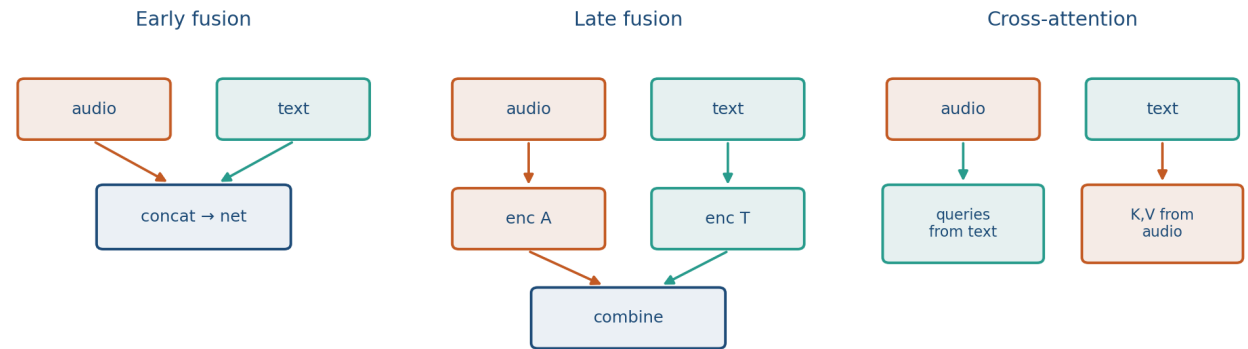


Early, late, cross-attention

- Early concat is simple and brittle if audio is missing.
- Cross-attention lets text query audio keys and values.

Three ways to mix streams. Missing audio at test time punishes early fusion.



Labeled audio is scarce

- Web text dwarfs image–text, which dwarfs transcribed audio.
- Transfer from CLIP/Whisper is the usual project move.

Paired, transcribed audio is the thin top of the data pyramid

